

Extract a Part from an Injection Mold Mock SOP (1x Episode: one part, one mold)

One episode takes **one** part out of the mold mock. Open the mold. Take the part out. Cut the gate off. Look at the gate. Put the part in the tray. Close the mold and set it up for the next part. The episode ends when the part lies in the tray, the runner is in the scrap cup, the cutters are back on the tool mat, and the mold is shut. Never split one part into two episodes.

The base is **passive**. It has no motor. You push it by hand to the bench in front of the mold mock and lock it there. Nothing is carried away to another table. Everything the episode touches is already there when recording starts: the mold mock bolted to the bench, the cutters on the tool mat, the tray, the scrap cup, and the reject bin.

The mold is used **as found**. It starts shut, with a part inside. The part is a molded plastic part with its runner still on it. The moving half slides on two guide bars. The episode ends with the mold shut and empty, ready for the next part to be loaded in the reset.

This is a bench task. Four rules come from that. First, the **mold mock stays bolted down**. You never push, pull, lift, or lean on the fixed half or the bench. Second, **the moving half only slides on its bars**. You pull it open by its handle until it stops, and you push it shut by its handle until it stops. You never twist it, lift it, or jerk it. Third, **the part face is never touched**. The face is the smooth, shiny side of the part. You lift the part out of the mold by its runner. After the cut you hold the part by its edges. Fourth, **the cutters only cut the gate**. They never touch the mold, the tray, or the face of the part.

Scrap goes in the scrap cup and nowhere else. The runner you cut off goes in the scrap cup. A bit of plastic left on the bench, in the mold, or on the floor is a loose bit. You do not chase a loose bit. You report it.

The order never changes: **open the mold; take the part out; cut the gate off; look at the gate; put the part in the tray; close the mold and reset it.**

Each gripper holds one thing at a time. Nothing is passed from one gripper to the other. The part is never in the air while the cutters move.

Setup

Do the base positioning and both checklists before you start an episode.

Cell configuration

- **cell_type:** bimanual

Base positioning

The base is **passive**. It has no motor. You push it into place by hand. You never steer it, nudge it, or move it after recording starts. You park it once, before recording, and it stays there until the episode is over.

1. Push the base by hand up to the bench. Stop it **square to the front edge of the bench**, so the front edge runs straight across the frame and both ends look the same distance away.
2. Stop it **in the middle of the mold mock**, so the tool mat side and the tray side are the same distance from the middle of the base.
3. Stop it **close enough** that a gripper reaches the runner in the open cavity without stretching the arm. Stop it **far enough** that no arm, wrist, or part of the base touches the bench front, the mold, or the guide bars while both arms work.
4. Check the **pull**. With the base parked, a gripper closes on the mold handle and slides the moving half all the way open and all the way shut without stretching.
5. Check the **cavity**. A gripper reaches the runner in the open cavity, lifts the part straight up, and clears the mold without stretching.
6. Check the **bench**. A gripper reaches the cutters on the tool mat, the tray, the scrap cup, and the reject bin. It does all four without stretching and without knocking anything over.
7. Lock the base. Push it once, hard, by hand. It must not roll, slide, or turn.
8. If any line from 1 to 6 fails, push the base to a new spot by hand and start again at line 1. Do not work at a bench the arms cannot reach easily.

The base stays locked and still for the whole episode. Nothing moves it. No arm leans on the bench hard enough to push it. Nobody touches it by hand. You never move it during the task. A base that moves after recording starts ends the episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera points at the middle of the mold mock. The frame shows the whole mold, both halves, the guide bars, the handle, the tool mat with the cutters, the tray, the scrap cup, and the reject bin.
3. The camera sees the parting line from the front, so you can see if the mold is open or shut.
4. The camera sees into the cavity when the mold is open, so you can see the part in it and see when it is empty.
5. The camera sees the tool mat from above, so you can see the part lying on it and the cutters on the gate.
6. The camera sees the gate on the part when it is held up at the inspection point, so you can see the cut.
7. The camera sees the tray pockets, so you can see where the part lands and which way its face points.
8. Both arms are at home with the grippers open.

9. The grippers reach the mold handle, the runner in the open cavity, the cutters, the tray, the scrap cup, and the reject bin without stretching to their limit.
10. The two arms do not hit each other when one holds the part still and the other holds the cutters.
11. If the arms cannot reach a place, park the base again with the Base positioning steps until lines 9 and 10 are all true.

Materials checklist

1. The **mold mock** is a two half mold made for practice. It does not heat up. Nothing is molded in it. A part is put in it by hand before each episode.
2. The **fixed half** is bolted to the bench. Nobody moves it, leans on it, or stands anything on it.
3. The **moving half** slides on two **guide bars**. It has a **handle** on its front. It stops hard at the open end and at the shut end.
4. The moving half slides easily. Test it by hand during setup. If it sticks, stop and fix the bars. Nobody forces it during the episode.
5. The **cavity** is the hollow in the fixed half. The part sits in it. When the mold opens, the part stays in the cavity of the fixed half.
6. The **part** is one molded plastic part. It has a smooth **face** that must not be scratched, and a **runner** stuck to one edge.
7. The **runner** is a plastic stick about as long as a finger. It is what you lift the part by.
8. The **gate** is the thin neck where the runner joins the part. It is the spot you cut.
9. The part sits loose in the cavity. It lifts out with no pull. Test it by hand during setup. If it sticks, change the part.
10. The **cutters** are flush cutters with two handles and a small flat jaw. They lie on the tool mat with the handles toward the front and the jaw pointing to the back.
11. The cutters are sharp and close all the way. Test them on a scrap runner during setup.
12. The **tool mat** is a rubber mat on the bench to one side of the mold. The cutters live on it. The part is laid on it for the cut. Nothing else is on it.
13. The **tray** is a flat tray with soft pockets. Each pocket holds one part face up. It stands on the bench on the other side of the mold with at least one empty pocket.
14. The **scrap cup** is an open cup next to the tray. The cut runner goes in it.
15. The **reject bin** is an open bin behind the tray. A part with a bad gate goes in it.
16. The bench top is clean and clear at the start. No loose bits of plastic lie on it, in the cavity, or on the tool mat.

Workspace layout

Nothing is marked or taped out. You judge every place below by eye, from the bench and the mold themselves.

- **Mold mock:** the middle of the bench. Fixed half at the back, moving half at the front, handle facing the front. Never moved.
- **Cavity:** the hollow in the fixed half. The part starts here and leaves here once.
- **Open end:** where the moving half stops when the mold is pulled open.
- **Shut end:** where the moving half stops when the mold is pushed shut.
- **Tool mat:** on one side of the mold. The cutters start and end here. The part is laid here for the cut.
- **Cut point:** the middle of the tool mat. The part lies here face up while the gate is cut, so the runner falls on the mat and not in the mold.
- **Inspection point:** the air above the tool mat, in front of the camera, where the gate is held up to be looked at.
- **Tray:** on the other side of the mold. Good parts go here, one to a pocket, face up.
- **Scrap cup:** next to the tray. The cut runner goes here.
- **Reject bin:** behind the tray. A part with a bad gate goes here.

Gripper rules

- **Handle:** one gripper on the handle, and only on the handle. It never touches the guide bars or the halves.
- **Part:** one gripper on the runner out of the mold, and on the edges after the cut. It never touches the face.
- **Cutters:** one gripper on the two handles of the cutters. The cutters are never in a gripper while the part is in the air.
- **Hold:** a closed gripper pressed on the edge of the part to keep it still on the mat. A hold never lifts or moves the part.
- Nothing is passed from one gripper to the other. Only one thing moves at a time.

Vocabulary

- **Parting line:** the gap between the two mold halves. You see it when the mold opens.
- **Open all the way:** the moving half stands at its open end stop. It does not move when you pull more.
- **Shut all the way:** the moving half stands at its shut end stop. The parting line is closed. No gap shows.
- **Runner grip:** the gripper closes on the runner in its middle, not on the gate and not on the part.
- **Edge grip:** the gripper closes on the part by two edges, with nothing touching the face.
- **Straight lift:** the part goes straight up out of the cavity, held by its runner, with no tilt and no drag along the cavity wall.
- **Low carry:** the part is carried close over the bench and level. It never goes over the mold.

- **Face up:** the smooth face of the part points at the ceiling.
- **Hold:** a closed gripper pressed on the edge of the part, so the part does not slide or spin on the mat.
- **Flush cut:** the cutters close right at the gate, with the flat side of the jaw against the part edge, so no stub sticks out.
- **Gate stub:** the small bit of the gate left on the part after the cut.
- **Good gate:** the stub is flat with the part edge or smaller than a grain of rice, and the part edge has no tear and no hole.
- **Bad gate:** the stub sticks out longer than a grain of rice, or the cut tore into the part, or there is a hole at the gate.
- **Seated in the pocket:** the part lies flat in one tray pocket, inside the pocket walls, face up, and does not rock.
- **Loose bit:** a piece of plastic that lands anywhere but in the scrap cup. You do not chase it. You report it.
- **Reset:** the mold is shut all the way, the cutters lie on the tool mat, the mat is clear, and both grippers are open and clear.
- **Back and clear:** the arm is pulled back, so no part of it is over the mold, the mat, or the tray.

Steps

Step 1: Open the mold

Goal: the moving half stands at its open end, and the part sits in the cavity of the fixed half.

1.1 Take the handle

- Close a gripper on the mold handle in the middle.
- Do not touch the guide bars, the fixed half, or the parting line.

Check: the gripper holds the handle square. Nothing else is touched. If the grip is off the middle, open it and take the handle again.

1.2 Slide it open

- Pull the handle straight toward the front, slow and level, along the guide bars.
- Do not twist it. Do not lift it. Do not jerk it.
- Pull until the moving half stops at the open end.

Check: the mold is **open all the way**. The parting line is a wide gap. The part sits in the cavity of the fixed half. If the part came out stuck to the moving half, stop and report it.

1.3 Let go of the handle

- Open the gripper and pull it back from the handle.

Expected state: the mold stands open. The part sits in the cavity. Both grippers are back and clear.

Step 2: Take the part out

Goal: the part lies face up on the tool mat at the cut point, with its runner sticking out to the side.

2.1 Take the runner

- Bring a gripper straight down over the cavity.
- Close on the runner in the middle. Do not close on the gate. Do not close on the part.

Check: it is a **runner grip**. The gripper holds only the runner. If it holds the part or the gate, open it and take the runner again.

2.2 Lift it straight up

- Lift the part straight up out of the cavity in a **straight lift**.
- Do not tilt it. Do not drag it along the cavity wall. Do not scrape it on the moving half.
- Lift until the part is clear of the mold.

Check: the part hangs from its runner, clear of the mold. The cavity is empty. If the part catches, lower it back into the cavity and lift again, slower.

2.3 Lay it on the tool mat

- Carry the part to the tool mat in a **low carry**. Do not carry it over the tray.
- Lower it straight down onto the middle of the mat, **face up**, with the runner pointing away from the cutters.
- Open the gripper and pull it back.

Check: the part lies flat on the mat, face up, and does not rock. The runner lies flat on the mat. If the part is face down, take the runner again, lift it, turn it, and lay it down again.

Expected state: the part lies face up on the tool mat. The cavity is empty. The mold is still open. The cutters have not moved.

Step 3: Cut the gate off

Goal: the runner is cut off at the gate with a flush cut, the runner is in the scrap cup, and the cutters are back on the mat.

3.1 Hold the part still

- Close a gripper and press it down on the edge of the part, on the side away from the runner.
- Press just hard enough that the part cannot slide or spin. Do not press on the face.

Check: it is a **hold**. The part does not move when pushed a little. The gripper is on the edge, not the face.

3.2 Pick up the cutters

- With the other gripper, close on the cutters by their two handles.
- Lift them just off the tool mat.

Check: the cutters hang jaw down, closed. Both handles are in the gripper. If the grip is on the jaw, put them down and take them again.

3.3 Set the jaw on the gate

- Open the cutters. Bring the jaw down to the gate.
- Lay the flat side of the jaw against the edge of the part, so the gate is between the blades.
- Do not touch the part face with the jaw. Do not put the runner in the jaw.

Check: the jaw sits on the gate, flat against the part edge, with nothing else between the blades. If the jaw is on the runner instead of the gate, open it and set it again.

3.4 Make the cut

- Close the cutters all the way in one squeeze.
- Keep the hold on the part while you cut.

Check: it is a **flush cut**. The runner lies loose on the mat. The part did not move. If the runner is still joined by a thread of plastic, set the jaw on the thread and cut again.

3.5 Put the cutters back

- Lift the cutters just off the part and carry them to their place on the mat.
- Lay them down with the handles toward the front and the jaw pointing to the back.
- Open the gripper and pull it back.

Check: the cutters lie closed on the mat where they started. If they lie on the part or off the mat, take them again and lay them down again.

3.6 Put the runner in the scrap cup

- Let go of the hold. Close a gripper on the loose runner.
- Lift it just off the mat and carry it to the scrap cup in a low carry.
- Open the gripper over the cup so the runner drops in.

Check: the runner is in the scrap cup. Nothing lies on the mat except the part and the cutters. If the runner fell on the bench, leave it and report it.

Expected state: the part lies face up on the mat with a gate stub. The runner is in the scrap cup. The cutters are back on the mat. Both grippers are back and clear.

Step 4: Look at the gate

Goal: you have seen the gate stub in front of the camera and know if it is good or bad.

4.1 Pick the part up by its edges

- Close a gripper on the part by two edges in an **edge grip**. Do not touch the face.
- Lift it just off the mat.

Check: the part hangs level in the gripper. Nothing touches the face. If the grip is on the face, put the part down and take it again.

4.2 Hold the gate up to the camera

- Lift the part to the **inspection point** above the mat.
- Turn it so the gate stub faces the camera. Hold it still for **2 seconds**.

Check: the gate stub is in the frame and in focus. You can see the stub, the part edge next to it, and any tear or hole.

4.3 Decide good or bad

- If the stub is flat or smaller than a grain of rice, and the edge has no tear and no hole, it is a **good gate**. Go to Step 5.
- If the stub sticks out longer than a grain of rice, or the edge is torn, or there is a hole, it is a **bad gate**. Carry the part in a low carry to the reject bin, open the gripper over the bin, and let it drop in. Then go to Step 6.
- Do not cut a bad gate again. Do not trim the stub.

Expected state: the part is either in the gripper, ready for the tray, or in the reject bin. The mat holds only the cutters.

Step 5: Put the part in the tray

Goal: the part lies face up in one empty tray pocket.

5.1 Carry it to the tray

- Carry the part in a **low carry** to the tray. Do not carry it over the mold.
- Bring it over the nearest empty pocket.

5.2 Lower it into the pocket

- Lower the part straight down into the pocket, **face up**.
- Open the gripper and pull it back.

Check: the part is **seated in the pocket**. It lies flat, inside the pocket walls, face up, and does not rock. If it sits on the pocket wall, or face down, take it by its edges, lift it, and put it down again.

Expected state: the part lies in a tray pocket. The mat holds only the cutters. The mold is still open and empty.

Step 6: Close the mold and reset it

Goal: the mold is shut all the way, and the workspace is ready for the next part.

6.1 Take the handle

- Close a gripper on the mold handle in the middle.
- Do not touch the guide bars, the fixed half, or the parting line.

6.2 Slide it shut

- Look into the cavity first. It must be empty.
- Push the handle straight toward the back, slow and level, along the guide bars.
- Do not twist it. Do not lift it. Do not slam it.
- Push until the moving half stops at the shut end.

Check: the mold is **shut all the way**. The parting line is closed and no gap shows. If a gap shows, pull the moving half open a little and push it shut again. If it will not shut, stop and report it.

6.3 Let go and check the bench

- Open the gripper and pull it back from the handle.
- Look at the tool mat. The cutters lie on it, closed, handles to the front. Nothing else lies on it.
- Look at the bench. No loose bits lie on it.

Check: the workspace is **reset**. The mold is shut, the cutters are in place, and the mat is clear. If the cutters are out of place, take them by their handles and lay them down again.

Expected state: the mold is shut and empty. The part is in the tray or in the reject bin. The runner is in the scrap cup. The cutters are on the mat. Both grippers are back and clear.

Step 7: End the episode

- Check the mold: it is shut all the way, with no gap at the parting line.
- Check the cavity: it is empty.
- Check the part: it lies face up in one tray pocket, or it is in the reject bin.
- Check the scrap cup: the runner is in it.
- Check the cutters: they lie closed on the tool mat, handles to the front.
- Check the bench: no loose bits lie on it.
- Check the base has not moved.
- Fix any failed check before you end.
- Send both arms home with the grippers open. Then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Pull the moving half open by hand.
2. Put a new part in the cavity by hand, with its runner on the side away from the cutters. Press it in so it sits flat.
3. Lift the part out by hand once and put it back, to check it comes out with no pull. Change the part if it sticks.
4. Push the moving half shut by hand until it stops.
5. Empty the scrap cup if it is more than half full.
6. Pick up any loose bits from the bench, the mat, the cavity, and the floor.
7. Check the cutters: they close all the way and cut a scrap runner clean. Change them if they crush instead of cut. Lay them on the mat, closed, handles to the front, jaw to the back.
8. Check the tray has at least one empty pocket. Change the tray if it is full.
9. Check the guide bars: the moving half slides open and shut by hand with no catch.
10. Check the fixed half is still bolted tight and does not rock.
11. Check the base is still locked and parked square. Then do the Base positioning steps and both Setup checklists again.

SOP violations

These are actions that break the SOP. They are reviewed side by side in the review tool.

How to record a violation in review

For each violation, write down the **start timestamp**, the **violation name**, and the **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining. It is not an annotation label.

Episode handling

Tag every violation with its timestamp and its name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Base moved during the episode

- **Visible cue:** the bench moves in the frame, the mold changes angle in the frame, or the base rolls, slides, or turns at any time after recording starts.
- **SOP rule broken:** Steps 1 to 7. The base is parked and locked before recording and stays still for the whole episode.

- **Coaching note:** park it, lock it, push it to test it, then start recording. A base that moves is a lost episode.

Violation: Mold half forced or lifted

- **Visible cue:** the moving half is twisted, lifted off the guide bars, jerked, or slammed. Or the handle is pulled sideways.
- **SOP rule broken:** Steps 1.2 and 6.2. Slide the moving half straight along its bars, slow and level, until it stops.
- **Coaching note:** the bars are the road. Straight along, until the stop, and no harder than that.

Violation: Fixed half touched or leaned on

- **Visible cue:** a gripper, wrist, or forearm rests on the fixed half, the guide bars, or the bench. Or the fixed half rocks.
- **SOP rule broken:** Steps 1 to 6. Nobody pushes, pulls, lifts, or leans on the fixed half or the bench.
- **Coaching note:** the arm holds itself up. If something moves when you touch it, you touched it too hard.

Violation: Held by the wrong part

- **Visible cue:** the gripper closes on the part face when lifting it out of the mold. Or it closes on the gate instead of the runner. Or it closes on the cutters by the jaw.
- **SOP rule broken:** Steps 2.1, 3.2, and 4.1. Lift by the runner, hold by the edges, and take the cutters by their handles.
- **Coaching note:** runner out of the mold, edges after the cut, handles on the cutters. The face is never touched.

Violation: Part dragged out of the cavity

- **Visible cue:** the part tilts, scrapes along the cavity wall, or catches on the moving half on the way out.
- **SOP rule broken:** Step 2.2. Lift the part straight up out of the cavity.
- **Coaching note:** straight up first. Nothing moves sideways until the part is clear of the mold.

Violation: Carried over the mold or the tray

- **Visible cue:** the part, the runner, or the cutters cross over the open mold, or the part crosses over the tray before Step 5. Or a thing is carried high instead of low and level.
- **SOP rule broken:** Steps 2.3, 3.6, 4.3, and 5.1. Every carry is low and level and never crosses the mold.
- **Coaching note:** low and around. Nothing crosses the open mold.

Violation: Part laid down wrong

- **Visible cue:** the part is laid on the mat face down, or off the mat, or with the runner under it. Or it is dropped instead of lowered.
- **SOP rule broken:** Step 2.3. Lay the part face up in the middle of the mat, with the runner flat and pointing away from the cutters.
- **Coaching note:** face up, runner out. A part you cannot see, you cannot cut.

Violation: Cut without a hold

- **Visible cue:** the cutters close on the gate while no gripper holds the part still. Or the part spins or slides on the mat during the cut.
- **SOP rule broken:** Steps 3.1 and 3.4. Hold the part still by its edge before and during the cut.
- **Coaching note:** hold first, cut second. A part that moves under the blades gets a torn gate.

Violation: Cut in the wrong place

- **Visible cue:** the cutters close on the runner away from the gate, leaving a long stub. Or the jaw is set on the part edge so the blades bite into the part.
- **SOP rule broken:** Steps 3.3 and 3.4. Set the flat side of the jaw against the part edge with the gate between the blades, and cut once.
- **Coaching note:** flat side to the part, gate between the blades. Not the runner, not the part.

Violation: Cutters touched the wrong thing

- **Visible cue:** the cutter jaw touches the part face, the mold, the tray, or the bench. Or the cutters are moved while the part is in the air.
- **SOP rule broken:** Steps 3.2 to 3.5. The cutters only cut the gate, and only while the part lies held on the mat.
- **Coaching note:** the cutters have one job. The part is on the mat when they come out and on the mat when they go back.

Violation: Cutters put back wrong

- **Visible cue:** the cutters are laid on the part, off the mat, open, or with the jaw to the front. Or they are dropped onto the mat.
- **SOP rule broken:** Step 3.5. Lay the cutters closed on the mat where they started, handles to the front and jaw to the back.
- **Coaching note:** back where they were, the same way round. The next episode starts from there.

Violation: Runner not put in the scrap cup

- **Visible cue:** the cut runner is left on the mat, on the bench, in the mold, or dropped on the floor. Or it is dropped into the tray or the reject bin.
- **SOP rule broken:** Step 3.6. The runner goes in the scrap cup and nowhere else.

- **Coaching note:** scrap has one home. Pick it up before the part comes off the mat.

Violation: Gate not shown to the camera

- **Visible cue:** the part goes from the mat to the tray with no stop at the inspection point. Or the gate stub is turned away from the camera. Or the hold is shorter than 2 seconds.
- **SOP rule broken:** Step 4.2. Hold the gate stub up to the camera, still, for 2 seconds.
- **Coaching note:** if the camera did not see the gate, nobody checked it.

Violation: Wrong call on the gate

- **Visible cue:** a part with a long stub, a tear, or a hole goes into the tray. Or a part with a good gate goes into the reject bin.
- **SOP rule broken:** Step 4.3. Good gate to the tray, bad gate to the reject bin.
- **Coaching note:** rice grain rule. Flat or smaller goes to the tray. Longer, torn, or holed goes to the bin.

Violation: Bad gate trimmed

- **Visible cue:** the cutters come back out after the inspection, and a second cut is made on the stub.
- **SOP rule broken:** Step 4.3. Do not cut a bad gate again. Do not trim the stub.
- **Coaching note:** one cut per part. A bad cut is a reject, not a retry.

Violation: Part not seated in the pocket

- **Visible cue:** the part ends up on a pocket wall, across two pockets, face down, rocking, or on top of another part.
- **SOP rule broken:** Step 5.2. Lower the part straight down into one empty pocket, face up, so it lies flat.
- **Coaching note:** straight down into one pocket, face up. Watch it settle before you let go.

Violation: Mold shut on something

- **Visible cue:** the moving half is pushed shut while the part, the runner, or a loose bit is still in the cavity or on the parting line.
- **SOP rule broken:** Step 6.2. Look into the cavity first. It must be empty before the mold shuts.
- **Coaching note:** look, then shut. A mold shut on plastic is a damaged mold.

Violation: Mold not shut all the way

- **Visible cue:** the episode ends with a gap at the parting line, or the moving half stopped short of the shut end.
- **SOP rule broken:** Step 6.2. Push until the moving half stops at the shut end and no gap shows.

- **Coaching note:** to the stop, then look for the gap. Almost shut is open.

Violation: Done in the wrong order

- **Visible cue:** the cutters come out before the part is on the mat. Or the part goes to the tray before the gate is shown to the camera. Or the mold is shut before the part is in the tray or the bin.
- **SOP rule broken:** Steps 1 to 6. Open, take out, cut, look, tray, close. One thing at a time.
- **Coaching note:** the order is the task. Skipping a step is not faster, it is a lost episode.

Violation: Something dropped or knocked over

- **Visible cue:** the part, the runner, or the cutters slip out of a gripper and fall on the bench or the floor. Or the scrap cup, the tray, or the reject bin is knocked over by an arm going past.
- **SOP rule broken:** Steps 1 to 6. One thing moves at a time, and each one is put down under control.
- **Coaching note:** close all the way before you lift, and put it down before you let go.

Violation: Loose bit chased

- **Visible cue:** a gripper goes after a bit of plastic on the bench, in the mold, or on the floor during the episode.
- **SOP rule broken:** Steps 1 to 6. You do not chase a loose bit. You report it.
- **Coaching note:** leave it, tag it, keep going. Clean up is not on camera.

Violation: Failed check not retried

- **Visible cue:** a check in a step clearly fails, and the episode carries on with no retry. For example, the part lies face down on the mat, the jaw is on the runner, or the part rocks in the pocket.
- **SOP rule broken:** Steps 1 to 6. Do the retry written under each failed check before you move on.
- **Coaching note:** a check is only worth doing if you do the retry. Fix it where it happened.

Violation: Wrong ending

- **Visible cue:** the episode ends with the mold open, the cavity not empty, the part not in the tray or the bin, the runner not in the cup, the cutters off the mat, an arm not home, or a gripper not fully open.
- **SOP rule broken:** Step 7. Check the mold, the cavity, the part, the scrap cup, the cutters, the bench, and the base. Then send both arms home with the grippers open and stop recording.
- **Coaching note:** check first. Going home is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, throw the episode away, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Camera cannot see the cavity, the gate stub, or the parting line**, so nobody can tell if the part came out clean, if the cut was good, or if the mold shut.
- **Hardware fault on an arm:** gripper failure, drift, a crash caused by the controller, or a motor error.
- **Base fault:** a lock that will not hold, or a base that rolls, slides, or turns with the lock on.
- **Faulty mold mock:** a moving half that sticks or binds on its bars, a stop that does not stop, a handle that comes loose, or a fixed half that rocks on its bolts.
- **Faulty part:** one that sticks in the cavity under a correct straight lift, one with no runner, one with a runner that snaps off in a correct grip, or one that arrives already cracked or scratched.
- **Faulty cutters:** ones that crush instead of cut, that will not close all the way, that come apart, or that are too big to fit on the gate.
- **Faulty tray:** pockets too small or too big for the part, or a tray that slides on the bench under a correct set down.
- **A place turns out to be too far for its arm** with the base parked correctly, so the handle, the runner in the cavity, the cutters, the tray, the scrap cup, or the reject bin cannot be reached without stretching or folding the arm.

Annotation subtasks (from SOP)

1. Take the mold handle
2. Slide the mold open
3. Take the runner in the cavity
4. Lift the part straight out of the cavity
5. Lay the part face up on the tool mat
6. Hold the part still on the mat
7. Pick up the cutters
8. Cut the gate
9. Put the cutters back on the mat
10. Put the runner in the scrap cup
11. Pick the part up by its edges
12. Hold the gate up to the camera
13. Put the part in the tray or the reject bin
14. Take the mold handle

15. Slide the mold shut
16. Send both arms home and end the episode